Multivariate Analysis

Continuous Assessment 10

Jadene Jaftha (JFTJAD003)
Isabella Lethbridge (LTHISA001)

10 May 2026

Question 1

a

```r
# Partition sample correlation matrix:
rho   <- corr_mat_profit
rho11 <- rho[1:6,1:6]
rho12 <- rho[1:6,7:8]
rho21 <- rho[7:8,1:6]
rho22 <- rho[7:8,7:8]

rho11_inv      <- solve(rho11)
eig11          <- eigen(rho11_inv)
sqrt_rho11_inv <- eig11$vectors%*%diag(sqrt(eig11$values))%*%t(eig11$vectors
   )

rho22_inv      <- solve(rho22)
eig22          <- eigen(rho22_inv)
sqrt_rho22_inv <- eig22$vectors%*%diag(sqrt(eig22$values))%*%t(eig22$vectors
   )

RHO <- sqrt_rho11_inv%*%rho12%*%rho22_inv%*%rho21%*%sqrt_rho11_inv

# Eigen decomposition:
eigRHO <- eigen(RHO)

# Obtain the canonical loadings:
a_1 <- sqrt_rho11_inv%*%eigRHO$vectors[,1]
a_1 <- (-1)*a_1
```

$$\hat{\mathbf{a}}_1 = \begin{bmatrix} 0.4581 \\ -0.1909 \\ -0.7009 \\ -2.7059 \\ 1.3417 \\ 0.9954 \end{bmatrix}$$

b

```r
b_1p <- rho22_inv%*%rho21%*%a_1

# Standardise to unit variance:
V1_var <- 1/(sqrt(t(b_1p)%*%rho22%*%b_1p))
V1_var <- matrix(c(V1_var, V1_var), 2, 1)
b_1    <- b_1p*(V1_var)
```

$$\hat{\mathbf{b}}_1(proportional) = \begin{bmatrix} -1.0370 \\ 0.3140 \end{bmatrix}$$

$$\hat{\mathbf{b}}_1 = \begin{bmatrix} -1.38120 \\ 0.4183 \end{bmatrix}$$

c

```
rho_1 <- sqrt(eigRHO$values[1])
```

$$\rho_1 = 0.7508$$

d

As variables 1, 5 and 6 increase $\boldsymbol{U}_1$ increases. Variable 4 has the largest effect on $\boldsymbol{U}_1$. $\boldsymbol{V}_1$ represents the difference between REV and Q.

e

```
A_z <- t(matrix(a_1))
B_z <- t(matrix(b_1))

corr_U1Z1 <- A_z%*%rho11
corr_V1Z2 <- B_z%*%rho22
```

$$\boldsymbol{\rho}_{(\boldsymbol{U}_1,\boldsymbol{X}^{(1)})} = \begin{bmatrix} -0.8133 & -0.4236 & -0.7214 & -0.8619 & -0.5747 & -0.7787 \end{bmatrix}$$

$$\boldsymbol{\rho}_{(\boldsymbol{V}_1,\boldsymbol{X}^{(2)})} = \begin{bmatrix} -0.9893 & -0.8759 \end{bmatrix}$$

All variables in $\boldsymbol{X}^{(1)}$ have a strong, negative correlation with $\boldsymbol{U}_1$. Variables 4 and 1 have the strongest correlation. Both market-value measure of profitability are highly correlated with $\boldsymbol{V}_1$.

Question 2

a

The aim of CCA is to maximise correlation between the two sets of variables.

$$Corr(\boldsymbol{X}_2^1, \boldsymbol{X}_1^2) = \frac{0.95}{\sqrt{1 * 1}} = 0.95,$$
$$Corr(\boldsymbol{X}_1^1, \boldsymbol{X}_2^2) = 0.$$

b

If $Corr(\boldsymbol{X}_2^1, \boldsymbol{X}_1^2) = 0$, there will be no correlation between the first canonical variables.

$$\begin{bmatrix} \Sigma_{11} & 0 \\ 0 & \Sigma 22 \end{bmatrix}$$

$$Cov(\boldsymbol{U}, \boldsymbol{V}) = \boldsymbol{a}^T \Sigma_{12} \boldsymbol{b} = 0,$$
$$Corr(\boldsymbol{U}, \boldsymbol{V}) = \frac{\boldsymbol{a}^T \Sigma_{12} \boldsymbol{b}}{\sqrt{\boldsymbol{a}^T \Sigma_{11} \boldsymbol{a}} \sqrt{\boldsymbol{b}^T \Sigma_{22} \boldsymbol{b}}} = 0.$$

c

$$Corr(\boldsymbol{X}_2^{(1)}, \boldsymbol{X}_1^{(2)}) = \frac{0.95}{\sqrt{2 * 2}} = 0.475.$$

Doubling the variances, increases the standard deviation by a factor of $\frac{1}{\sqrt{2}}$. The denominator becomes larger while correlation stays the same, hence the canonical correlation decreases. The first canonical correlation decreases from 0.95 to 0.475.

d

We initially had $\mathbf{a} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, and the variances as unit variances:

$$Var(\boldsymbol{U}_1) = \boldsymbol{a}^T \Sigma_{11} \boldsymbol{a} = 1$$
$$Var(\boldsymbol{V}_1) = \boldsymbol{b}^T \Sigma_{22} \boldsymbol{b} = 1.$$

$$\begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 100 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 1$$

and

$$\begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 100 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 1$$

After doubling the variances:

$$\begin{bmatrix} 0 & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 100 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix} = 1$$

and

$$\begin{bmatrix} \frac{1}{\sqrt{2}} & 0 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 100 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix} = 1$$

The lengths of the first canonical coefficient vectors decrease to ensure that $Var(\boldsymbol{U}_1) = 1$ and $Var(\boldsymbol{V}_1) = 1$.

Question 3

For random vectors $\boldsymbol{X}^{(1)}$ and $\boldsymbol{X}^{(2)}$, the first canonical coefficient vectors $\boldsymbol{a}, \boldsymbol{b}$ maximise

$$Cor(\boldsymbol{a}^T\boldsymbol{X}^{(1)}, \boldsymbol{b}^T\boldsymbol{X}^{(2)} = \boldsymbol{a}^T\boldsymbol{\rho}_{12}\boldsymbol{b}.$$

where $\boldsymbol{\rho}_{12}$ is the correlation matrix between $\boldsymbol{X}^{(1)}$ and $\boldsymbol{X}^{(2)}$.

$$\begin{aligned} \boldsymbol{Z}^{(1)} &= \boldsymbol{V}_{11}^{-1}\boldsymbol{X}^{(1)}, \\ \therefore \boldsymbol{X}^{(1)} &= \boldsymbol{V}_{11}\boldsymbol{Z}^{(1)}. \end{aligned}$$

and

$$\begin{aligned} \boldsymbol{Z}^{(2)} &= \boldsymbol{V}_{22}^{-1}\boldsymbol{X}^{(2)} \\ \therefore \boldsymbol{X}^{(2)} &= \boldsymbol{V}_{22}\boldsymbol{Z}^{(2)} \end{aligned}$$

Thus, we have:

$$\boldsymbol{U} = \boldsymbol{a}^T\boldsymbol{X}^{(1)} = \boldsymbol{a}^T\boldsymbol{V}_{11}\boldsymbol{Z}^{(1)} = (\boldsymbol{V}_{11}\boldsymbol{a})^T\boldsymbol{Z}^{(1)}.$$

and

$$\boldsymbol{V} = \boldsymbol{b}^T\boldsymbol{X}^{(2)} = \boldsymbol{b}^T\boldsymbol{V}_{22}\boldsymbol{Z}^{(2)} = (\boldsymbol{V}_{22}\boldsymbol{b})^T\boldsymbol{Z}^{(2)}.$$

The corresponding first canonical coefficient vectors $\boldsymbol{a}^z, \boldsymbol{b}^z$ maximise:

$$Cor(\boldsymbol{a}^{zT}\boldsymbol{X}^{(1)}, \boldsymbol{b}^{zT}\boldsymbol{X}^{(2)}) = \boldsymbol{a}^{zT}\boldsymbol{\rho}_{12}^z\boldsymbol{b}^z = \boldsymbol{a}^{zT}\boldsymbol{\rho}_{12}\boldsymbol{b}^z.$$

We know that $\boldsymbol{\rho}_{12}^z = \boldsymbol{\rho}_{12}$ and we have $Var(\boldsymbol{U}_k) = 1$ and $Var(\boldsymbol{V}_k) = 1$.

Since the canonical variables are constrained to have unit variance, the coefficients need to change to ensure this holds.

Hence,

$$\begin{aligned} \boldsymbol{a}^z &= \boldsymbol{V}_{11}\boldsymbol{a} \\ \boldsymbol{b}^z &= \boldsymbol{V}_{22}\boldsymbol{b} \end{aligned}$$

The canonical coefficients must change because they are not scale invariant (unlike the canonical correlations).